

Sampling and Observational vs. Experimental Data

QBA 5330 – Module 1, Part 2 (September 5)

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We framed a decision in Part 1.

Now: where does the data actually come from — and can we trust it?

Population vs. Sample

Example: a retailer wants average satisfaction across 40,000 loyalty members.

- **Population** — all 40,000 members
- **Sample** — the members actually surveyed
- **Sampling frame** — the list you actually draw from (e.g., members with a valid email)

Three Sampling Methods

- **Simple random sampling** — every member has an equal chance
- **Stratified sampling** — split into subgroups (e.g., tier), sample within each
- **Convenience sampling** — survey whoever is easiest to reach

A Line Worth Remembering

A convenience sample answers “what do the people who were easy to reach think?” — which may or may not be the question you wanted answered.

Observational Data

We **watch and record** what naturally happens, without intervening.

Example: comparing revenue at stores that happened to stay open late on their own, vs. stores that didn't.

Experimental Data

We **deliberately assign** an intervention and observe the result.

Example: randomly assigning some stores to stay open late, and comparing to stores that weren't assigned to.

The Core Message

Only well-designed experiments, with **random assignment**, let us make confident **causal** claims.

Observational comparisons are often **confounded** – some other difference between groups may be driving the result.

Does a Wellness Program Reduce Sick Days?

Observational: compare sick days between employees who *chose* to enroll and those who didn't.

- Problem: enrollees may already be healthier or more conscientious

Experimental: *randomly assign* employees to the program or a control group.

- Groups should be comparable on everything else, *on average*

Activity: Diagnose the Study

For your scenario, answer:

1. Observational or experimental?
2. What sampling method (if any)?
3. What could go wrong with trusting the result?

Before the Next Break

We now know where trustworthy data comes from.

Next: turning it into something we can actually look at — tables, graphs, and summary numbers.